CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 3 From the 461 Final

Solutions

Only the 461 final questions that fall within the midterm

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01 Solution Q1: MCQ

#	Answer	Reason
3	**c. Collision**	The definition of a collision: different elements mapped to the same cell.
4	**b**	Chaining does not avoid a collision. It handles one after it happens. The other three reduce the chance of it happening.

02 Solution Q2: short answers

A) Divide, then conquer, then combine: break the problem into sub-problems of the same form, solve them recursively, and combine the solutions.

B) Merge sort and quick sort.

03 Solution Q3: analysis

Line	# operations	Why
`1. w ← 0`	`1`	An assignment.
`2. c ← 1`	`1`	An assignment.
`3. while w < n² do`	`n² + 1`	The loop runs `n²` times, and the test runs once more when it fails.
`4. w ← w + 1`	`2n²`	An addition and an assignment.
`5. c ← c * a`	`2n²`	A multiplication and an assignment.
`6. return c`	`1`	The return.
Total	**`5n² + 4`**	`O(n²)`

$$T(n) = 1 + 1 + (n^2 + 1) + 2n^2 + 2n^2 + 1 = 5n^2 + 4 \quad \rightarrow \quad O(n^2)$$

The algorithm multiplies `c` by `a` a total of `n²` times, so it returns `a` to the power `n²`. It has no separate best and worst cases, because the number of iterations depends on `n` only.

If you count computing `n²` in every test as an extra operation, the total becomes `6n² + 5`. The result is the same: `O(n²)`.